LEDGER · 2026-09-01 · POLYMER CHEMISTRY / OILFIELD CHEMICALS

Lignosulfonate-Acrylamide Graft Copolymers (LS-g-PAAm) via Redox-Initiated Aqueous Polymerization

REJECTED

This concept did **not** clear the framework.

The reasoning is published in full below.

What the assessment found

Incumbent benchmark	Driscal D (Chevron Phillips Chemical)
Entry application	HTHP fluid loss control additive in water-based drilling fluids for deep oil, gas, and geothermal wells
Measured advantage	3.84x reduction in HTHP fluid loss at 200°C (7.5 mL for LS-g-PAAm vs 28.8 mL for Driscal D)
Time to first revenue	6 months
Gross margin at parity	87.7%
Startup CapEx	$92,000
Payback from first sale	0.6 months
Capital productivity	39.13x

Regulatory risk	Moderate — qualification costs reported ($65,000 estimated)

Assessment

The venture proposal meets all seven criteria and passes all sub-tests. It demonstrates a 3.84x reduction in HTHP fluid loss compared to the market leader (Driscal D) at 200°C while achieving 88% gross margins at price parity. Startup CapEx is $92,000 with a 6-month timeline to initial revenue.

Also flagged

- Evidence rests on non-peer-reviewed sources
- Duplicate references inflating the evidence base
- Price parity asserted, not sourced
- no_production_runsheet

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-01 against seven gates plus the voiding sub-tests. How the method works · Browse all concepts